Engineering design for a circular economy

Investigating the design process

The design strategy most relevant to our project was circular product design. During this project week, we took into account the 4 R's while designing the functionality of our robotic arm. We considered how the arm might be reused after the project week. Our arm can be controlled manually by the user using two joysticks. This broadens the use case from just the project week, as the arm is not just constrained to moving between fixed positions. As the arm is controlled by an Arduino, future users could reprogram it for additional functionality as well.

Impact of LAC (CO_2 Footprint)

Raw materials
The arm is mostly aluminium and lexan polycarbonate. The CO_2 cost of lexan is approximately 4.5kg of CO_2 per kg of lexan. The mass of lexan in the arm is approximately 270g of lexan, giving a mass of CO_2 of 1.2kg. The cost of aluminium production in Europe is approximately 7kg of CO_2 per kg of aluminium. The mass of aluminium in the arm is approximately 135g, giving a mass of CO_2 of 0.945.
Transport
Applying the transport CO_2 data from the "Carbon Footprint Analysis of Printed Circuit Board" to the SSC-32 board on the arm, which has an area of 0.0045m^2 gives the CO_2 cost of transportation of the board as 8.28kg of CO_2. Applying the same data to the Arduino, as it is likely produced in a very similar way, gives a CO_2 cost of 2.57kg.
Manufacturing
Again, this time applying the manufacturing data from the "Environmental Carbon Footprints" paper gives the CO_2 cost of the SSC-32 board to be 0.2kg.
There is no publicly available information for the CO_2 impact of the ESP-32 Arduino Nano we used in the design but applying the same documentation before, using the dimensions of the board (0.0014 m^2) gives a CO_2 cost of 0.06kg for the entire board.
Product Lifespan
During the lifespan of the arm the main CO_2 output comes from its need for power. The power of the arm comes to 30Wh. The average mass of CO_2 emitted per kWh in the UK is 308g. Assuming our product is used for an hour daily, this gives a mass of emitted CO_2 of 3.4kg per year. For the purposes of calculating an overall impact, we will assume the product has a lifespan of 5 years, giving an approximate CO_2 output of 16.8kg.

End-of-life
Assuming that the unrecyclable parts of the product are put into landfill, then the CO_2 cost of those parts will be negligible. From transport and processing of the aluminium, using the weight from before, gives a CO_2 of 0.4kg for the process.

An estimate for the total CO_2 impact can be found by simply adding the CO_2 outputs from above, giving an overall CO_2 output of 31.96kg. For cost saving, we can sell the SSC-32, Arduino and other components as E-waste. The impact of reusing our product is that it can stay in the product lifespan stage for longer, instead of being broken down or replaced by a new arm, which has already had a costly CO_2 impact in its creation. The impact of recycling materials is that it takes less CO_2 to recycle them than to extract more.

Monetary value created

As mentioned above, following the recycle R, we can sell the electronics of the arm as E-waste to make some money back from disposal of the product. We could also sell the aluminium to be recycled. It would likely not be a lot of money (approximately 40p per arm for the electronics and 80p for the aluminium.)